Data Communication

Content 07: Error Correction & Detection

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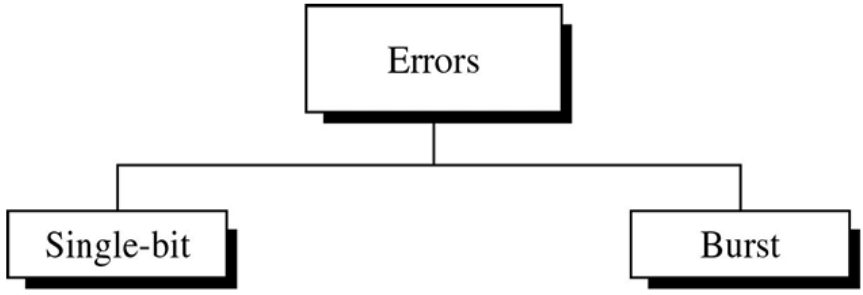
Error Detection & Correction

- **Detection**: looking only to see if any error has occurred.
- **Correction**: need to know the exact number of bits that are corrupted and more importantly, their location in the message.
- The correction of errors is more difficult than the detection.
- Error Detection and Correction are implemented at the data link layer and the transport layer.

Error Detection & Correction

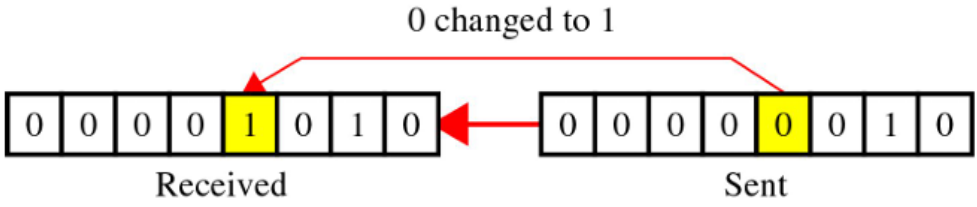
- **Forward Error Correction**
  The process in which the receiver tries to guess the message by using redundant bits.
- **Retransmission**
  The technique in which the receiver detects the occurrence of an error and asks the sender to resend the message.

Types of Errors



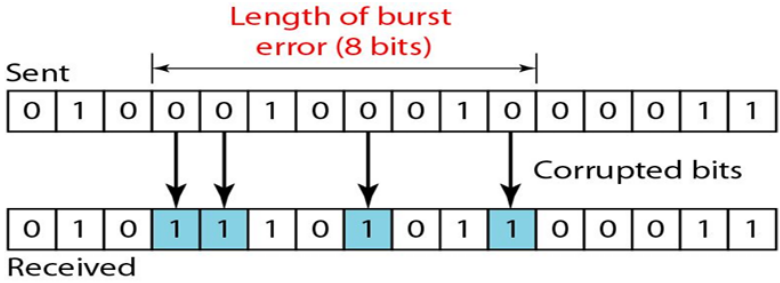
Single-Bit Error

Only 1 bit in the data unit has changed.



Burst-Bit Error

- A burst error means that two or more bits in the data unit have changed.
- The length of the burst is measured from the first corrupted bit to the last corrupted bit. Some bits in between may not have been corrupted.

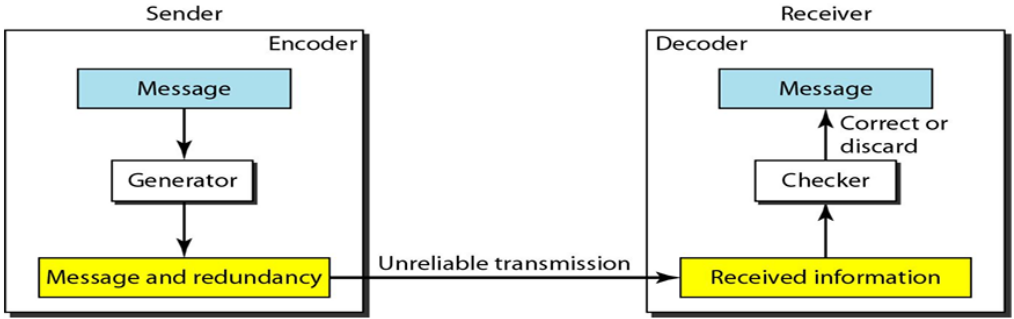


Redundancy

- The central concept in detecting or correcting errors is Redundancy.
- To detect or correct errors, we need to send extra (redundant) bits with data. This technique is called **Redundancy**.
- These extra bits are discarded as soon as the accuracy of the transmission has been determined.
- Redundancy is achieved through various coding schemes.

Coding Schemes

- The sender adds redundant bits through a process that creates a relationship between redundant bits and the actual data bits.
- The receiver checks the relationships between the two sets of bits to detect or correct the errors.



Error Detection Schemes

As there are many error detection schemes, we will focus on a few fundamental and widely used ones in this course:

- **Single Parity Check**
- **Two-Dimensional Parity Check**
- **Checksum**
- **Cyclic Redundancy Check (CRC)**

Single-Parity Check

- Adds one extra bit (parity bit) to a data unit.
- Ensures total number of 1s is even (even parity) or odd (odd parity).
- **Example:**
  - **Even Parity**
    - Data: 1010101 (4 ones) $\Rightarrow$ Parity bit = 0 (even parity)
    - Transmitted: 10101010
  - **Odd Parity**
    - Data: 1010101 (4 ones) $\Rightarrow$ Parity bit = 1 (odd parity)
    - Transmitted: 10101011
- **Can detect:** All single-bit errors.
- **Cannot detect:** All burst errors (if the number of flipped bits is even).

Single-Parity Check: Examples (Even Parity)

- **Example 01**
  - Data: `1010101` (4 ones)
  - Parity Bit: `0` (to keep count of 1s even)
  - Transmitted: `10101010`
  - Received: `10100010` $\Rightarrow$ Even number of 1s $\Rightarrow$ **No error**
- **Example 02**
  - Transmitted: `10101010`
  - Suppose a bit is flipped: `11101010`
  - Received: 5 ones (odd) $\Rightarrow$ **Error detected**
- **Example 03**
  - Transmitted: `10101010`
  - Suppose two bits are flipped: `11101110`
  - Received: 6 ones (even) $\Rightarrow$ **Error not detected!**

Two-Dimensional Parity Check

- Improves error detection by adding parity bits to both rows and columns.
- Can detect and correct single-bit errors.

Original data: 10 11

```
1 0
1 1
```

After adding row and column parity (even parity):

```
1 0 | 1
1 1 | 0
-------------
0 1 | 1
```

Final Data: 10 11 101 011

Two-Dimensional Parity: Example

Original Data: 1011 0101 1110 0011

```
1 0 1 1
0 1 0 1
1 1 1 0
0 0 1 1
```

Adding parity bits: (even parity)

```
 1 0 1 1 | 1
 0 1 0 1 | 0
 1 1 1 0 | 1
 0 0 1 1 | 0
-------------------
 0 0 1 1 | 0
```

Final Data: 1011 0101 1110 0011 10100 00110

Two-Dimensional Parity: Error Detection

- **Single-bit errors:** Easily detected — row and column parity both mismatch.
- **Two-bit errors:**
  - If the two bits are in *different rows and columns*, both row and column parities will mismatch $\rightarrow$ detected.
  - If the two errors occur in the *same row and same column*, the parities may cancel out $\rightarrow$ not detected.
- **Three-bit errors:**
  - Always change an odd number of row and/or column parities.
  - Cannot "cancel out" completely $\rightarrow$ will be detected.
- **Limitation:**
  - 2D parity can detect (but not correct) up to 3-bit errors.
  - Cannot detect all 4-bit errors or even some 2-bit errors.

Checksum

- Sender and receiver agree on a block size (e.g., 4 bits, 8 bits, etc.).
- Data is divided into equal segments and summed.
- The sender transmits the complement of this sum.
- The receiver verifies by repeating the process; mismatches indicate errors.

Analogy:
Data: (7, 3) $\rightarrow$ Sum = 10 $\rightarrow$ Checksum = -10
Send: (7, 3, -10) $\rightarrow$ Total = 0 = No Error

One's Complement Arithmetic

- **One's complement** flips each bit:
  - 0 becomes 1, 1 becomes 0.
  - Example: $1010 \rightarrow 0101$
- **One's Complement Addition:**
  - Add binary numbers normally.
  - If a carry is generated, **wrap it around** and add to the least significant bit.
- **Example:**
  - Add: $1101 + 1011 = 11000 \rightarrow$ drop carry bit: $1000 + 1 = 1001$
  - Final sum $= 1001$
- **Checksum:** Take 1's complement of this sum.

Checksum: Example

Original Data: 1001 0101 1100 0011

```
1001
0101
1100
0011
```

Step 1: Add all blocks using 1's complement addition

```
1001 + 0101 = 1110
1110 + 1100 = 1 1010 → wrap around: 1010 + 1 = 1011
1011 + 0011 = 1110
```

Step 2: Take 1's complement of 1110 → Checksum = 0001
Final Data: 1001 0101 1100 0011 0001

Checksum: Example

Verification of Received Data: 1001 0101 1100 0011 0001

```
1001
0101
1100
0011
0001 (checksum)
```

Step 1: Add all blocks with checksum

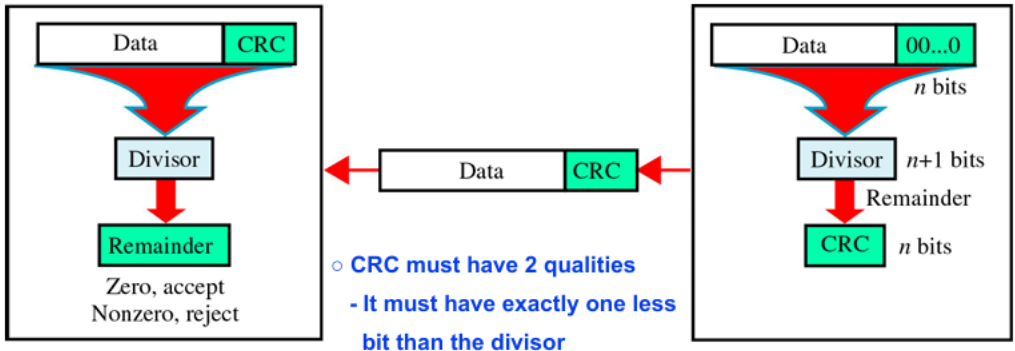
```
Same as before: Final sum = 1110
1110 + 0001 = 1111
```

Step 2: Take 1's complement of 1111 → 0000 → No error

Cyclic Redundancy Check (CRC)

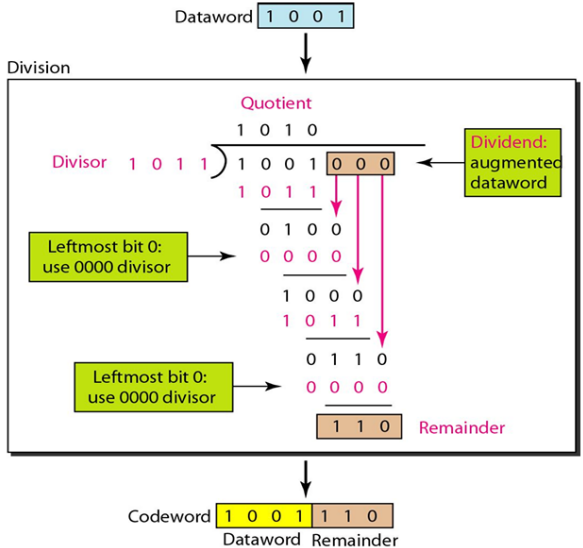
- CRC (Cyclic Redundancy Check) is a powerful error detection technique.
- It treats data as a polynomial and divides (binary XOR) it by a fixed generator polynomial.
- The remainder from this division is the *CRC code*.
- The CRC checksum is appended to the data before transmission.
- At the receiver, data + CRC is divided by the same polynomial.
- If remainder $= 0$, data is assumed error-free; else error detected.
- CRC can detect burst errors and multiple bit errors better than parity or checksum.

CRC

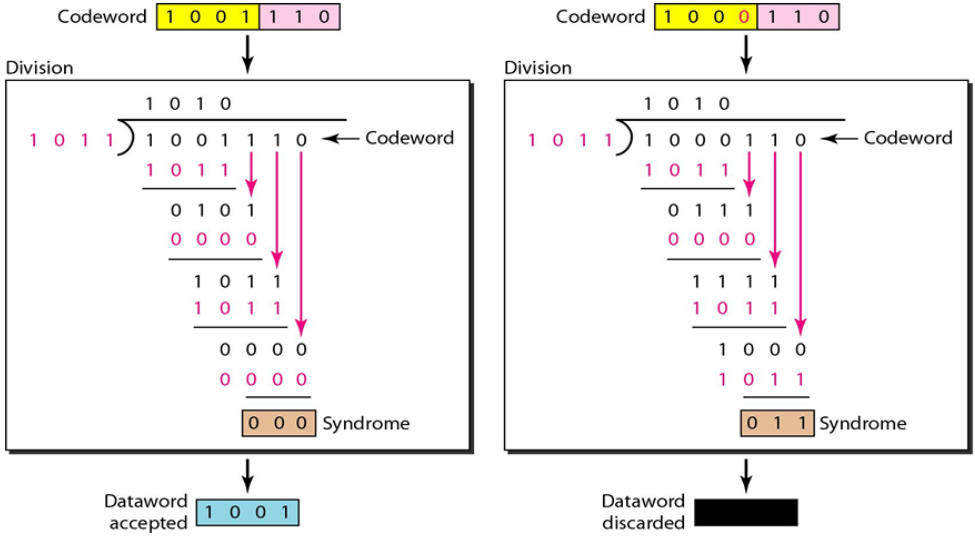


- CRC must have 2 qualities
  - It must have exactly one less bit than the divisor
  - Resulting bit sequence (Data+CRC) exactly divisible by the divisor.

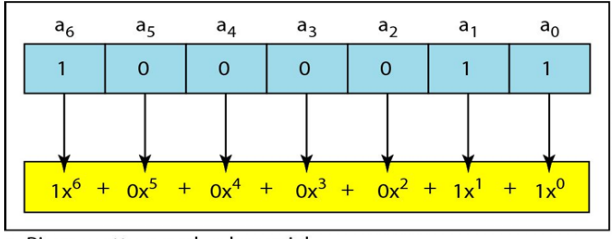
CRC: Example


Dataword 1 0 0 1
Division
Quotient
1 0 1 0
Divisor 1 0 1 1
1 0 0 1 0 0 0
Dividend: augmented dataword
1 0 1 1
0 1 0 0
0 0 0 0
Leftmost bit 0: use 0000 divisor
1 0 0 0
1 0 1 1
0 1 1 0
0 0 0 0
Leftmost bit 0: use 0000 divisor
1 1 0
Remainder
Codeword 1 0 0 1 1 1 0
Dataword Remainder

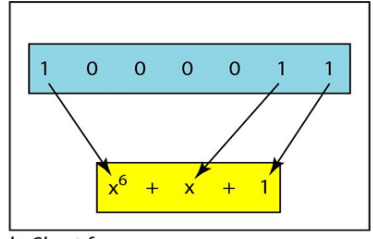
CRC: Example


Codeword 1 0 0 1 1 1 0
Division
1 0 1 0
1 0 1 1) 1 0 0 1 1 1 0 ← Codeword
1 0 1 1
0 1 0 1
0 0 0 0
1 0 1 1
1 0 1 1
0 0 0 0
0 0 0 0
0 0 0 Syndrome
Dataword accepted 1 0 0 1
Codeword 1 0 0 0 1 1 0
Division
1 0 1 0
1 0 1 1) 1 0 0 0 1 1 0 ← Codeword
1 0 1 1
0 1 1 1
0 0 0 0
1 1 1 1
1 0 1 1
1 0 0 0
1 0 1 1
0 1 1 Syndrome
Dataword discarded

CRC: Polynomial Representation



a. Binary pattern and polynomial



b. Short form

CRC: Standard

Name	*Polynomial*	*Application*
CRC-8	$x^8 + x^2 + x + 1$	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	LANs

CRC

Advantages:

- Very good performance
- Easily be implementation in hardware and software
- Faster

Error Correction: Hamming Code

- Error detection identifies data corruption during transmission or storage.
- Error correction goes a step further: fixes the error without retransmission.
- Hamming Code is an error correction scheme that:
  - Detects and corrects single-bit errors.
  - Locates the error position using parity bits.

Hamming Code: Example

- Hamming code C(7,4) adds parity bits to 4 data bits to make a 7-bit codeword.
- Bit positions: 1-based index – `7 6 5 4 3 2 1`
- Number of parity bits, m, is determined by:

$$2^m \geq m + d + 1$$

- For $d = 4$, minimum m satisfying the formula is $m = 3$:

$$2^3 = 8 \geq 3 + 4 + 1 = 8$$

- Parity bits at positions: 1, 2, 4 (powers of 2)
- Data bits placed in positions: 3, 5, 6, 7
- Positions: $[d_1, d_2, d_3, p_3, d_4, p_2, p_1]$

Hamming Code: Example

- Parity bits are calculated based on binary representation of positions.

Position	Binary	Covered by
1	001	p_1
2	010	p_2
3	011	p_1, p_2
4	100	p_3
5	101	p_1, p_3
6	110	p_2, p_3
7	111	p_1, p_2, p_3

$$p_1 = b_3 \oplus b_5 \oplus b_7$$
$$p_2 = b_3 \oplus b_6 \oplus b_7$$
$$p_3 = b_5 \oplus b_6 \oplus b_7$$

Hamming Code: Example

Example: Data = 1010

- Place data bits: `1 0 1 _ 0 _ _`
- Compute parity bits:

$$p_1 = b_3 \oplus b_5 \oplus b_7 = 1 \oplus 1 \oplus 1 = 1$$
$$p_2 = b_3 \oplus b_6 \oplus b_7 = 1 \oplus 0 \oplus 1 = 0$$
$$p_3 = b_5 \oplus b_6 \oplus b_7 = 1 \oplus 0 \oplus 1 = 0$$

Encoded message: `1 0 1 0 0 0 1`

(XOR calculation is as same as calculating even parity)

Hamming Code: Example

[Assuming no error]

- Received message: `1 0 1 0 0 0 1`
- Recalculate parity checks:

$$S_1 = b_1 \oplus b_3 \oplus b_5 \oplus b_7 = 1 \oplus 1 \oplus 1 \oplus 1 = 0$$
$$S_2 = b_2 \oplus b_3 \oplus b_6 \oplus b_7 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$
$$S_3 = b_4 \oplus b_5 \oplus b_6 \oplus b_7 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$

- Syndrome $= S_3S_2S_1 = 000_2 = 0 \rightarrow$ No Error
- Extract data bits: positions 3,5,6,7 $\rightarrow$ `1 0 1 0`

Hamming Code: Example

[Assuming an error in 3^{rd} bit]

- Received message: `1 0 0 0 0 0 1`
- Recalculate parity checks:

$$S_1 = b_1 \oplus b_3 \oplus b_5 \oplus b_7 = 1 \oplus 1 \oplus 0 \oplus 1 = 1$$
$$S_2 = b_2 \oplus b_3 \oplus b_6 \oplus b_7 = 0 \oplus 1 \oplus 0 \oplus 1 = 0$$
$$S_3 = b_4 \oplus b_5 \oplus b_6 \oplus b_7 = 0 \oplus 0 \oplus 0 \oplus 1 = 1$$

- Syndrome $= S_3S_2S_1 = 101_2 = 5 \rightarrow$ Error in bit 5
- Flip bit 5: Corrected $=$ `1 0 1 0 0 0 1`
- Extract data bits: positions 3,5,6,7 $\rightarrow$ `1 0 1 0`

References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 10: 10.1 - 10.5

Thank You